A Project Report

On

Transistor Level Implementation and Validation of a Switching Activity Optimised Reversible Full Adder

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR, ANANTAPUR

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In

ELECTRONICS & COMMUNICATION ENGINEERING

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DECLARATION

We hereby declare that the results embodied in this project **"Transistor Level Implementation and Validation of a Switching Activity Optimised Reversible Full Adder"** by us under the guidance of **Dr. R. Ravindraiah, Ph.D. Assistant Professor, Department of ECE** in partial fulfillment of the award of **Bachelor of Technology** in **Electronics and Communication Engineering, MITS, Madanapalle** from **Jawaharlal Nehru Technological University Anantapur, Anantapur** and we have not submitted the same to any other University/institute for award of any other degree.

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ABSTRACT

Power dissipation is a major challenge in modern VLSI systems due to aggressive technology scaling, higher integration density, and increased operating frequencies. In CMOS digital circuits, dynamic power caused by switching activity forms a significant portion of total power consumption. Reversible logic provides an attractive method for designing low-power digital systems because it has the potential to remove unnecessary signal transitions and eliminate data loss.

There are several reversible full adders that use different techniques to reduce the number of signal transitions that occur during switching activity. However, most of these designs were only tested on either the logic level or FPGA level. Therefore, this paper presents a reversible full adder implemented at the transistor level and tested using GPDK 90 nm CMOS technology. It also compares a design based on FG, TG, and NG gates to test the feasibility of implementing a reversible full adder at the transistor level. Due to the fact that direct CMOS implementation of the NG gate produces some logical inconsistencies due to the inversion issues present in previous implementations of the NG gate, we develop new implementations of the NG gate which can be used as replacements in previously developed optimal reversible full adder designs. then performed extensive transient simulations using Cadence Virtuoso to validate the functionality of the redesigned NG gate along with FG and TG. The simulation results indicate that the proposed reversible full adder uses less average power than its traditional non-reversible counterpart, yet maintains competitive delay characteristics when both designs are fabricated under the same process technology.

Keywords: Reversible logic, switching activity reduction, full adder, transistor-level design, low-power VLSI, CMOS circuits, GPDK 90 nm, Cadence Virtuoso.

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List of Abbreviations

S. No.	Abbreviation	Full Form
1	ADE	Analog Design Environment
2	AND	Logical AND Gate
3	CMOS	Complementary Metal-Oxide Semiconductor
4	CL	Load Capacitance
5	EDP	Energy Delay Product
6	FA	Full Adder
7	FG	Feynman Gate
8	FPGA	Field Programmable Gate Array
9	GPDK	Generic Process Design Kit
10	NG	New Gate
11	NMOS	N-type Metal-Oxide Semiconductor
12	NOR	NOT OR Gate
13	NOT	Logical Inverter
14	OR	Logical OR Gate
15	Pavg	Average Power
16	PDP	Power Delay Product
17	PMOS	P-type Metal-Oxide Semiconductor
18	TG	Toffoli Gate
19	VLSI	Very Large-Scale Integration
20	XOR	Exclusive OR Gate
21	α	Switching Activity Factor
22	f	Operating Frequency
23	tpd	Propagation Delay

CHAPTER 1
INTRODUCTION

1.1 Motivation

The rapid advancement of VLSI technology has enabled the development of highly complex and high-performance digital systems. However, as technology scales down to deep submicron levels, power consumption has emerged as a critical challenge in modern integrated circuit design. Excessive power dissipation not only reduces battery life in portable devices but also leads to thermal issues and reliability degradation in high-performance systems [1]. In CMOS digital circuits, a major portion of power consumption is attributed to dynamic power, which is primarily caused by switching activity during logic transitions. Each time there is a transition in a signal (that is, it goes from "logic '0'" to "logic '1'", or vice versa) there will be a use of power when it charges or discharges the load capacitance [2].

The greater the complexity of the circuit and/or the faster the operating frequency of the circuit, then the more transitions there will be in signals, and thus the larger the amount of energy consumed through switching activities. Traditional low-power design techniques such as reducing supply voltages, reducing clock frequencies, gate-clocking (i.e., turning off clocks), and transistor sizing have all become popular means for designers to decrease power consumption [3]. However, these traditional techniques are being less effective because they depend on continued reductions in CMOS technology dimensions.

Therefore, new design techniques need to be developed to significantly reduce power consumption. Reversible logic is becoming increasingly recognized as a viable method to reduce power consumption at the circuit level. In reversible logic, every input vector is mapped directly to exactly one output vector. Thus, reversible logic ensures that information is never destroyed during processing. Since Landauer showed that destroying one bit of information requires dissipating energy; reversible logic offers a way to minimize this dissipated energy [4].

Furthermore, Bennett showed that reversible logic could theoretically eliminate all energy losses associated with the destruction of information [5]. Full Adders are basic building blocks of most modern digital systems including microprocessors, digital signal processors, and Arithmetic Logic Units. Since they are so ubiquitous, decreasing their power consumption can lead to substantial improvements in overall system efficiency.

1.2 Problem Definition

Though reversible logic has great potential in the area of low-power design, there are some challenges that will have to be overcome before it can become a practical reality. Many existing reversible full adder designs were developed at the logic level or via an FPGA based approach; both methods fail to accurately model how a reversible full adder would operate at the transistor level of a CMOS circuit [6].

When reversible logic circuits are designed and then implemented with CMOS technology, designers run into issues related to polarity ambiguity, incorrect signal transition timing, and functionality mismatches. For example, there are reversible gates commonly found in full adder designs that cannot functionally produce valid output signals when they are implemented at the transistor level because the logic definition of those reversible gates does not correctly match the operational characteristics of transistors. While previous researchers typically have focused on theoretical aspects (i.e., gate count, number of garbage outputs, quantum cost), very few researchers have also considered the practical performance metrics of interest (e.g., power consumption, propagation delay, energy efficiency) for reversible logic circuits at the transistor level.

Therefore, there exists a need to create and verify reversible logic circuits at the transistor level to assure their operational correctness and performance under typical CMOS operating conditions. This research effort will address this challenge by developing a reversible full adder circuit optimized for reduced switching activity and validating its operation through transistor level simulation.

1.3 Objectives of the Project

The overall goal of this project is to create and test an ultra-low-power reversible full adder that operates at the transistor-level based upon CMOS technology. The specific goals of this project are as follows:

- To study the fundamentals of reversible logic and switching activity in digital circuits. design and verify basic CMOS logic gates using Cadence Virtuoso
- To implement reversible gates such as Feynman Gate (FG), Toffoli Gate (TG), and New Gate (NG) using transistor-level CMOS circuits
- To analyze the behavior of the NG gate and identify inconsistencies between logic-level representation and transistor-level implementation

- To redesign the NG gate based on its derived truth table to ensure correct functionality. To construct a switching-activity optimized reversible full adder using the corrected reversible gates
- To evaluate the performance of the proposed design in terms of power consumption, propagation delay, Power Delay Product (PDP), and Energy Delay Product (EDP). To compare the proposed reversible full adder with a conventional CMOS full adder and existing designs

1.4 Limitations of the Project

Although the proposed work provides a detailed transistor-level implementation of a reversible full adder, certain limitations exist:

- The design is implemented using a single technology node (90 nm CMOS), and performance variations across other technologies are not explored
- Physical layout design and post-layout analysis are not included in this work
- The implementation is limited to a 1-bit full adder and is not extended to multi-bit arithmetic circuits
- Although other factors (i.e. process variation, parasitic capacitance, etc.) were considered, they were not extensively examined.
- The primary focus of the research will be on reducing the amount of power consumed through switching activities, whereas leakage power consumption was not thoroughly studied.

1.5 Organization of Documentation

The remainder of this thesis is organized as follows:

- **Chapter 2** presents the literature survey, including existing reversible logic designs and switching activity reduction techniques
- **Chapter 3** provides the analysis of the system, including theoretical concepts, hardware and software requirements, and design methodology
- **Chapter 4** describes the design of the proposed system, including transistor-level implementation of reversible gates and the corrected NG gate
- **Chapter 5** presents the implementation results and performance analysis of the proposed reversible full adder
- **Chapter 6** discusses the testing and validation of the design, including verification using truth tables and simulation results
- **Chapter 7** concludes the work and summarizes the key contributions of the project

CHAPTER 2

LITERATURE SURVEY

2. 1 Introduction

There has been much research conducted over the past few decades in the area of low-power VLSI Design with an emphasis placed on reversible logic and the resultant switching activity. In addition, the subject matter presented here includes a review of prior art relevant to reversible logic, switching activity, and low-power circuit design.

As devices continue to shrink in size, so too does the dynamic power consumption associated with switching activity become larger than static power consumption. These trends have caused designers to look into different approaches to design low power circuits. One approach that is being researched is the use of reversible logic for low power designs [1]. Since reversible logic provides a 1:1 relationship between input(s) and output(s), it is possible to eliminate information loss from computations using reversible logic [2]. Theoretical studies have provided additional support for the use of reversible logic. Subsequent work has identified reversible computing as an attractive method for designing energy efficient systems [3-4]

2. 2 Related Works

The basis for reversible logic is found in Toffoli's [5] initial development of reversible computational models. Conservative logic concepts for reversible circuits were further developed by Fredkin and Toffoli [6]. Kerntopf [8] presented heuristic approaches to the design of reversible logic synthesizers. Quantum implementations of reversible arithmetic operations have also been investigated by Vedral et al. [9]. Additionally, reversible arithmetic circuits (including ALUs) were presented by Thapliyal and Ranganathan [10]. A wider scope of reversible logic design techniques was given by Rice [11]; reversible logic unit implementations were further explored by Pakala and Ranganathan [12].

Reversible logic low-power full-adder designs were demonstrated by Biswas et al. [13]. Important metrics that determine the performance of reversible circuits were discussed by Mohammadi and Eshghi [14]. The primary metric for evaluating the effectiveness of reversible logic circuits is the amount of switching activity present within those circuits. Muzio and Baillie [15] investigated and compared various techniques used to estimate the switching activity generated in digital systems. Benini et al. [16] studied how hazards affect the energy consumed by digital systems. Chandrakasan et al. [17] introduced CMOS-based design strategies to minimize the power consumed by digital systems; Roy et al. [18] examined leakage current in deep-submicron digital technology. Weste et al. [19-21] provide detailed descriptions of standard CMOS based design principles. Other works explore additional system level power minimization strategies.

Thapliyal et al. [22], have presented an efficient arithmetic circuit design for reversible logic. Reversible full adder designs using FPGAs were introduced by Rashid and Hasan [23]. Low power reversible adder designs were introduced by Sayedsalehi et al. [24]. Reversible logic synthesis was presented by Wille et al. [25], where reversible logic is synthesized through output permutations. A comprehensive review of reversible circuit optimizations was presented by Saeedi and Markov [26]. Technology mappings for reversible circuits are reviewed by Mishchenko et al. [27]. Sequential reversible circuits were studied by Angizi et al. [28]; Navi et al. [29] presented reversible full adder with optimized structure; and Wang et al. [30] researched minimizing switching activity in CMOS design.

2. 3 Disadvantages of Existing works

Although advances have been made, current reversible logic architectures suffer from multiple problems:

- Current reversible logic architecture is primarily tested for operation at either the logic or FPGA level; however, it does not adequately represent transistor-level CMOS behavior.
- Although many of these architectures focus on theoretically measured metrics like number of gates and quantum cost, they generally do not consider practically important metrics like power or delay.
- Techniques used to reduce switching activities are frequently executed at higher levels than circuit levels and therefore lack circuit-level validation.
- Directly mapping reversible logic to CMOS circuits can result in functional inconsistencies.
- Very little research has focused on transistor-level verification based upon real world CMOS technology.

2. 4 Proposed System

This paper will address some of the issues of previous works by focusing on transistor-level implementations and validations of reversible full adders that minimize switching activity. This proposed design uses reversible gates including Feynman Gates (FG) Toffoli Gates (TG) and New Gates (NG). However, during the implementation process there were discrepancies in the functioning of the NG gate relative to what was expected.

Therefore, the Truth Table for the NG was developed and analyzed, and subsequently redesigned to function properly. The redesigned NG gate is combined with additional reversible gates to develop the full adder. The full adder is then designed utilizing CMOS technology and its performance is examined through power consumption, delay, Power Delay Product (PDP), and Energy Delay Product (EDP). The results show an improvement over prior reversible full adder design.

2. 5 Advantages over Proposed System

The proposed design has a number of benefits compared with previous reversible full adder systems and other full adder designs:

- Practical Implementation at Transistor Level: The proposed full adder is designed and implemented at the transistor level utilizing CMOS technology; therefore, there should be no issue as to its practicability.
- Correct Functional Verification: Each circuit was functionally verified by means of simulation to ensure that they work correctly for all possible inputs.
- Resolution of NG Gate Issues: Anomalies present in the NG gate were recognized and corrected by designing an alternative version of the gate.
- Minimization of Power Consumption: The optimization of switching activity results in less dynamic power consumption than prior versions.
- Improvements in Energy Efficiency: This full adder design demonstrates improvements in both PDP (Power Dissipated Per Operation) and EDP (Energy Dissipated per Operation) when compared to prior designs.
- Realistic Simulation Conditions: The performance of this design was evaluated with respect to power consumption and delay under real-world simulation conditions.
- Use of a Structured Design Methodology: The use of a structured methodology, which provides correctness at each step in the design process, ensured the accuracy of this design.

The proposed design closes the gap between theoretical reversible logic designs and the practicality of implementing these designs in CMOS technology, thereby providing a viable solution for low power VLSI applications.

CHAPTER 3
ANALYSIS

3. 1 Introduction

In addition to the hardware and software requirements needed for implementing the switching activity optimized reversible full adder system discussed in this Chapter, the design process and validation of the proposed system are also presented. The focus of this Chapter will be on detailing the resources that need to be allocated by designers in order to develop and implement the proposed reversible full adder system, and describe the sequential steps involved in developing the proposed reversible full adder system using CMOS technology. A systematic method of designing and verifying the proposed reversible full adder system was implemented to ensure correct development and verification.

3. 2 Hardware & Software Requirement Specification

Implementation and validation of the proposed reversible full adder system were completed using simulation-based design. As described previously, the hardware and software requirements for this project include the following:

- **Hardware Requirements**

  Hardware requirements for this project were minimal since the simulation tools provided the ability to simulate the design. However, an adequate amount of computational power is necessary for efficient simulation.

  i. **Processor:** Intel Core i5 or higher
  ii. **RAM:** Minimum 8 GB (16 GB recommended)
  iii. **Storage:** At least 20 GB free space
  iv. **Operating System:** Linux / Windows compatible with Cadence tools

- **Software Requirements**

  The following software tools were utilized during the design and simulation phases of the proposed system:

  - Cadence Virtuoso: Utilized for creating schematics, simulating designs, and analyzing waveforms.
  - GPDK 90nm CMOS Technology Library: Provided transistor models for CMOS implementation.
  - Analog Design Environment (ADE): Utilized for transient simulations and performance analysis.

Cadence Virtuoso is one of the most commonly used EDA platforms in VLSI design for transistor level circuits implementations and verifications. With Cadence Virtuoso, designers can evaluate accurately how their transistors behave in real-world applications.

3. 3 Methodology

A structured bottom-up approach has been followed in developing the design to provide correct functionality at every stage of the design and implementation.

- **Methodology of Proposed System**

The methodological approach adopted for this project is as follows:

- **i.** Study of the base paper and reversible logic fundamentals
- **ii.** Design of basic CMOS logic gates based on transistors
- **iii.** Transient simulations to verify operation of basic logic gates
- **iv.** Generation of symbols for use in hierarchical design
- **v.** Development of reversible gates (FG, TG, NG)
- **vi.** Detection of a malfunctioning NG gate during simulations
- **vii.** Derivation of truth table by analyzing logic diagrams
- **viii.** Derivation of truth table by generating logic equations
- **ix.** Comparison of derived truth tables to determine inconsistencies
- **x.** Re-design and modification of NG gate to resolve issues
- **xi.** Construction of a reversible full-adder circuit utilizing corrected gates
- **xii.** Performance testing and evaluation via simulation

- **Flowchart of Proposed System**

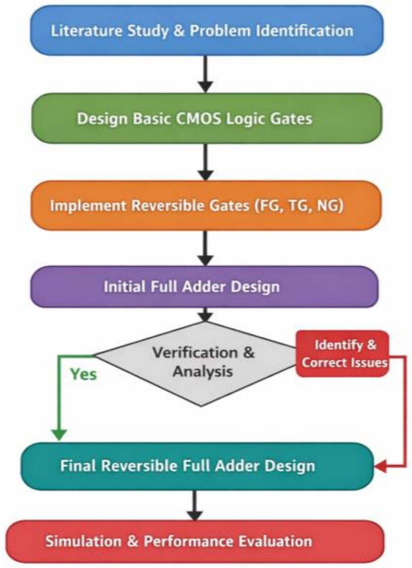

Figure 3.1: Flowchart of Proposed System

The flow chart shown in figure 3.1 represents the entire design process which was followed in this research. It begins with an examination of reversible logic fundamentals, then moves through transistor-level design, transient simulations and finally hierarchical modeling. The flow chart also demonstrates how the inconsistencies that were identified in the NG gate are addressed through the modifications made to it. Ultimately, the modified gates were utilized to create the reversible full-adder and test its performance by means of simulation. Through this structure and organization; a step-by-step approach is provided to develop and validate the system developed within this project.

- **Theoretical Background**

i. Reversible Logic

Conventional digital circuits are unable to recover all of the information contained within the inputs due to the inherent nature of processing signals and therefore lose some of that signal as they process it, resulting in energy loss. To solve the problem of losing information as a result of processing, reversible logic was developed. Unlike traditional digital circuits, reversible logic does not allow for two different input combinations to produce the same output combination. Therefore, there is no loss of information in reversible logic systems. Energy loss occurs when information is lost (according to Landauer's Principle) [1], but Bennett demonstrated that reversible computing could be used to theoretically eliminate that energy loss [2] in computational processes. While actual reversible circuits will never be able to completely stop energy dissipation from occurring, reversible logic allows us to limit switching activity and lower the amount of power required for operation.

ii. Reversible Logic Gates

Reversible logic circuits are constructed using special gates that preserve information. In this work, three reversible gates are used.

(a) Feynman Gate (FG)

The Feynman Gate is a 2×2 reversible gate used for signal copying and XOR operations.

$$P = A \tag{1}$$

$$Q = A \oplus B \tag{2}$$

where $\oplus$denotes the XOR operation.

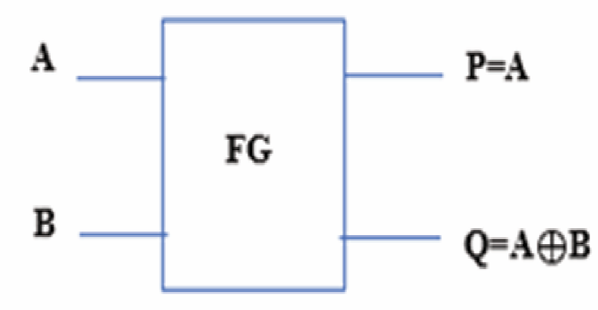

Figure 3.2: Symbol of FG

The figure 3.2 shows the symbolic representation of the Feynman Gate, which performs signal duplication and XOR operations.

Table 3.1: Truth table of FG

A	*B*	*P*	*Q*
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

The corresponding truth table 3.1 demonstrates the relationship between inputs and outputs, where one output replicates the input and the other performs XOR operation. This gate is widely used in reversible logic circuits for copying signals without violating reversibility constraints.

(b) Toffoli Gate (TG)

The Toffoli Gate is a 3×3 reversible gate used for control-based operations.

$$P = A \tag{3}$$

$$Q = B \tag{4}$$

$$R = (A \cdot B) \oplus C \tag{5}$$

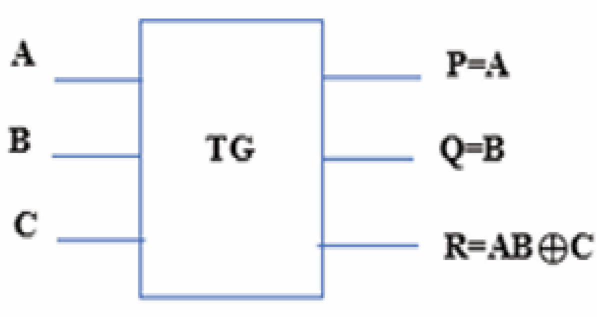

Figure 3.3: Symbol of TG

Figure 3.3 illustrates a reversible Toffoli Gate, which is a universal reversible gate capable of being used to create numerous types of logic operations. This gate is very powerful because it can be used to implement AND operations in reversible logic circuits.

Table 3.2: Truth table of TG

A	***B***	***C***	***P***	***Q***	***R***
0	*0*	*0*	*0*	*0*	*0*
0	*0*	*1*	*0*	*0*	*1*
0	*1*	*0*	*0*	*1*	*0*
0	*1*	*1*	*0*	*1*	*1*
1	*0*	*0*	*1*	*0*	*0*
1	*0*	*1*	*1*	*0*	*1*
1	*1*	*0*	*1*	*1*	*1*
1	*1*	*1*	*1*	*1*	*0*

The table 3.2 demonstrates how the third output can depend upon the control inputs and enable conditional operations. The reversible Toffoli gate plays an important role in creating reversible digital arithmetic circuits.

(c) New Gate (NG)

The New Gate is a 3×3 reversible gate used in switching-activity optimized full adder design.

$$P = A \tag{6}$$

$$Q = A \oplus B \tag{7}$$

$$R = (A \cdot B) \oplus C \tag{8}$$

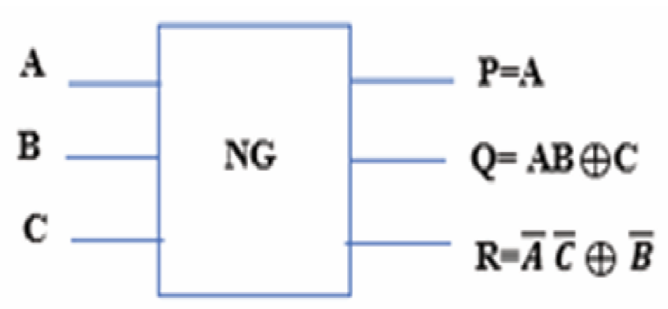

Figure 3.4: Symbol of NG

The figure 3.4 shows the symbolic representation of the New Gate (NG), and the corresponding truth table 3.3 defines its input-output relationship.

Table 3.3: Truth table of NG

A	B	C	P	Q	R
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

It is important to note that the implementation of the NG gate based on the design reported in [31] may lead to inconsistencies at the transistor level. Therefore, in this work, the NG gate is analyzed and redesigned to ensure correct functionality.

iii. Switching Activity

Switching activity refers to the frequency of transitions between logic '0' and '1' in a digital circuit. Each switching event results in charging and discharging of capacitive nodes, leading to dynamic power consumption.

The dynamic power consumption in CMOS circuits is given by:

$$P_{dynamic} = \alpha C_L V_{DD}^2 f \tag{9}$$

where:

- α= switching activity factor
- C_L= load capacitance
- V_{DD}= supply voltage
- f= operating frequency

Reducing switching activity directly reduces dynamic power consumption.

iv. Power Dissipation in CMOS

Power dissipation in CMOS circuits consists of two main components:

- **Static Power:** Caused by leakage current, which increases with technology scaling
- **Dynamic Power:** Caused by switching activity and is the dominant component

In this work, the focus is primarily on reducing dynamic power consumption.

v. Propagation Delay

Propagation delay is the time required for the output of a circuit to respond to a change in input.

$$t_{pd} = t_{output} - t_{input} \tag{10}$$

Lower propagation delay indicates faster circuit operation.

vi. Performance Metrics

To evaluate the performance of the proposed design, the following metrics are used:

Power Delay Product (PDP)

$$PDP = P_{avg} \times t_{pd} \tag{11}$$

PDP represents the energy consumed per switching operation.

Energy Delay Product (EDP)

$$EDP = PDP \times t_{pd} \tag{12}$$

EDP evaluates both energy efficiency and speed of the circuit.

vii. Reversible Full Adder Concept

A full adder is a combinational circuit that performs the addition of three input bits: A, B, and C_{in}. The outputs are Sum and Carry. In reversible logic, the full adder is constructed using reversible gates such as FG, TG, and NG while maintaining a one-to-one mapping between inputs and outputs.

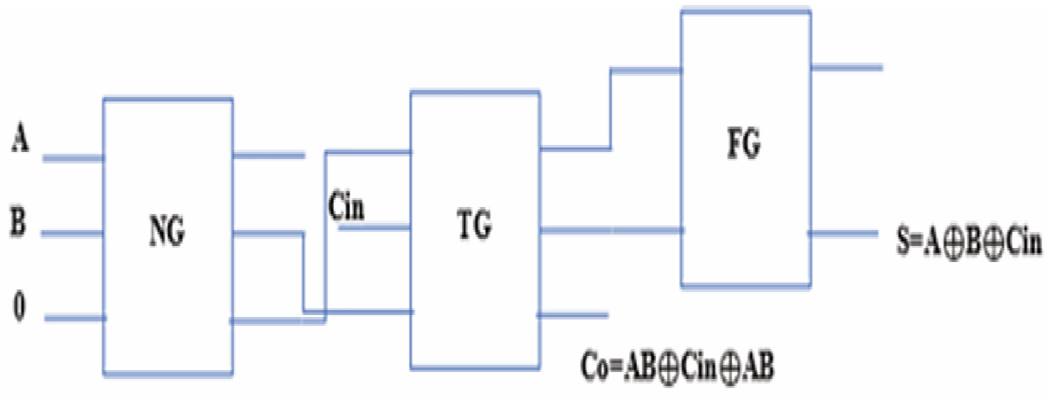

Figure 3.5: Block diagram of full adder using reversible gates

The figure 3.5 illustrates the structure of the reversible full adder constructed using FG, TG, and NG gates. The inputs are processed through reversible gates to generate Sum and Carry outputs without loss of information. The design focuses on minimizing switching activity and ensuring correct functional behavior.

3. 4 Summary

This chapter presented the theoretical background and analysis required for the proposed design. Key concepts such as reversible logic, switching activity, CMOS power dissipation, and performance metrics were discussed. The reversible gates used in the design were also introduced. These concepts provide the foundation for the transistor-level implementation described in the next chapter.

CHAPTER 4
DESIGN

4. 1 Introduction

This chapter presents the detailed design of the proposed switching-activity optimized reversible full adder at the transistor level. The design follows a systematic bottom-up methodology, starting from basic CMOS logic gates and progressing towards the implementation of reversible gates and the complete full adder circuit.

The objective of this design is to achieve correct functionality while minimizing switching activity and power consumption. During the implementation process, inconsistencies were identified in the New Gate (NG), which were further analyzed and corrected. The entire design is implemented using Cadence Virtuoso with GPDK 90 nm CMOS technology.

4. 2 Block Diagrams

This section presents the overall design flow and architecture of the proposed system.

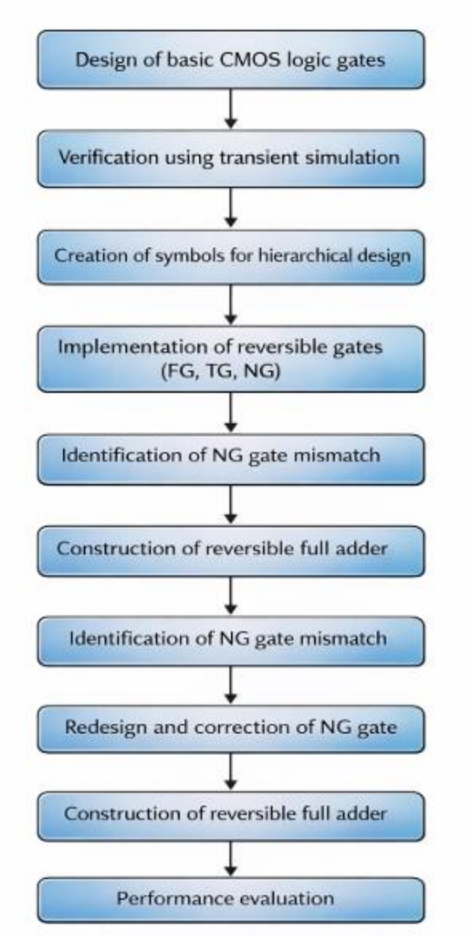

Figure 4.1: Overall Design Flow of Proposed System

The complete design process that was followed in this research is illustrated with Figure 4.1. In the beginning, it begins with the design of the fundamental CMOS logic gates. Then, they have been checked (verified) and symbolic representations were made. Verified modules are used to create reversible gates, which will be subsequently assembled into reversible full adders. Also, the process has included identifying and correcting inconsistencies in the NG gate. The use of a structured methodology provides an orderly and trustworthy method of developing and implementing the proposed system.

4. 3 Module Design and Organization

The proposed design utilizes a hierarchical methodology. Initially, basic CMOS logic gates are designed and verified. These gates are then used to construct reversible gates, which are further integrated to build the complete reversible full adder.

- **CMOS Logic Gate Design**

  The design begins with the implementation of basic CMOS logic gates, which act as building blocks for reversible circuits. A bottom-up approach ensures that each gate is verified before being used in higher-level designs.

  CMOS Inverter (NOT Gate)

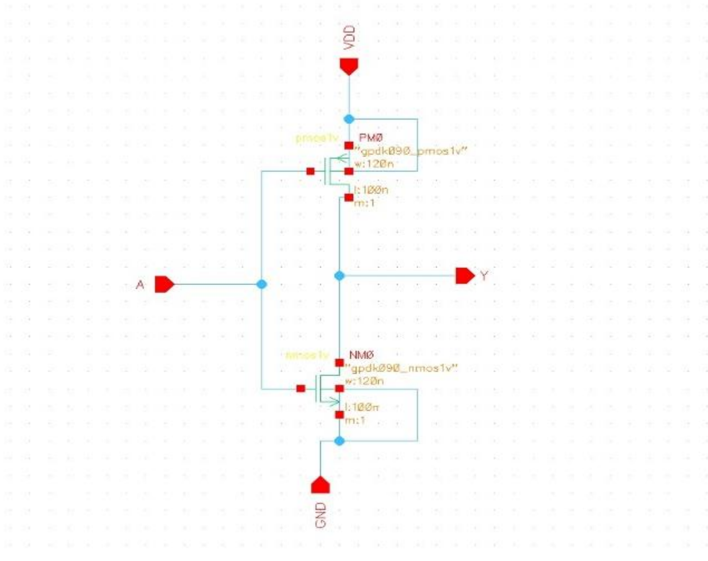

Figure 4.2: Transistor-level schematic of CMOS inverter

The figure 4.2 shows the transistor-level schematic of the CMOS inverter implemented using one PMOS and one NMOS transistor. The complementary operation of these transistors ensures low power consumption and proper switching behavior. When the input is low, the PMOS conducts and produces a high output, whereas when the input is high, the NMOS conducts and produces a low output.

CMOS NAND Gate

The CMOS NAND gate is one of the fundamental universal logic gates and is widely used in digital circuit design.

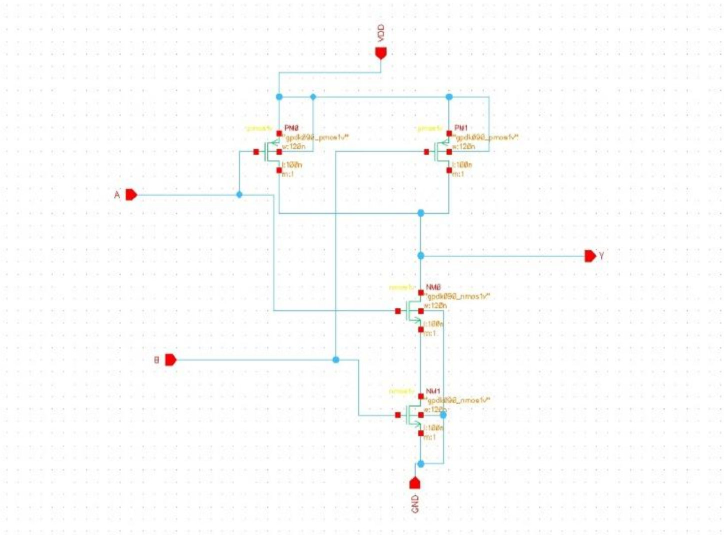

Figure 4.3: Transistor-level schematic of CMOS NAND gate

The figure 4.3 illustrates the CMOS NAND gate implemented using series-connected NMOS transistors and parallel-connected PMOS transistors. The output becomes low only when both inputs are high, otherwise it remains high.

CMOS NOR Gate

This gate is also a universal gate and plays an important role in logic circuit design.

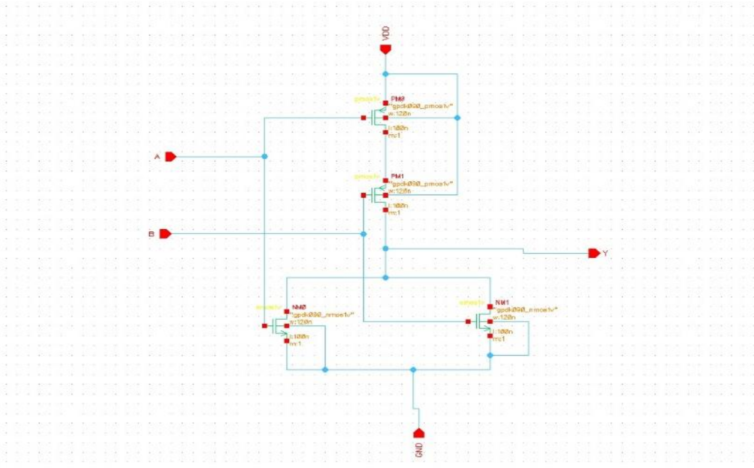

Figure 4.4: Transistor-level schematic of CMOS NOR gate

The figure 4.4 shows the CMOS NOR gate where NMOS transistors are connected in parallel and PMOS transistors are connected in series. The output is high only when all inputs are low. When any input is high, the NMOS network conducts and pulls the output to ground.

CMOS AND Gate

The CMOS AND gate is implemented by combining a NAND gate followed by an inverter.

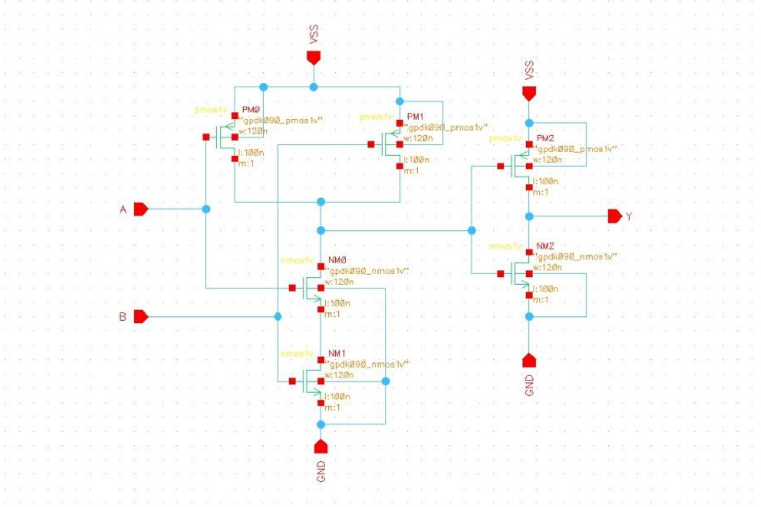

Figure 4.5: Transistor-level schematic of CMOS AND gate

The figure 4.5 demonstrates how the NAND output is inverted to obtain the AND functionality. This ensures correct logical operation using standard CMOS structures.

CMOS OR Gate

The CMOS OR gate is realized using a NOR gate followed by an inverter.

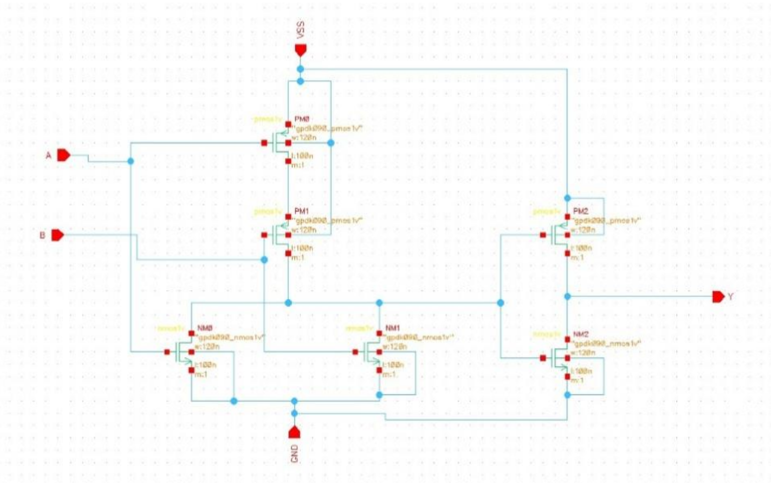

Figure 4.6: Transistor-level schematic of CMOS OR gate

The figure 4.6 shows how the NOR gate output is inverted to produce OR gate functionality. This implementation ensures efficient and reliable logic realization.

- **Verification and Symbol Creation**

All logic gates are verified using transient simulation in Cadence Virtuoso. Each module in the hierarchy was individually verified for each input combination to validate that each module performed as designed. Once validated:

- Symbols were generated
- Stored in a library
- Used in hierarchical design

➢ **Implementation of Reversible Gates**

Verified CMOS logic gate symbols were used to develop reversible logic gates.

Feynman Gate (FG)

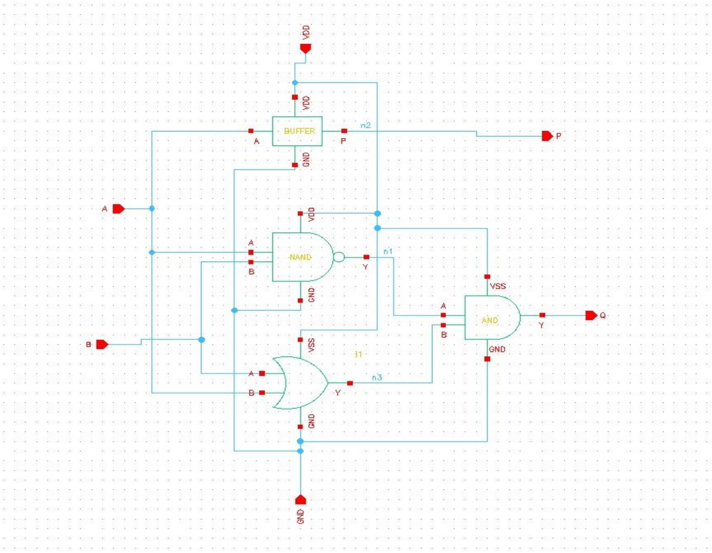

Figure 4.7: Transistor-level schematic of FG

Figure 4.7 provides an illustration of the Feynman gate at a transistor level implemented with CMOS logic. It utilizes standard CMOS logic components to achieve XOR and copy functions. As described above, it was built from CMOS logic components whose correct logical function has been previously validated.

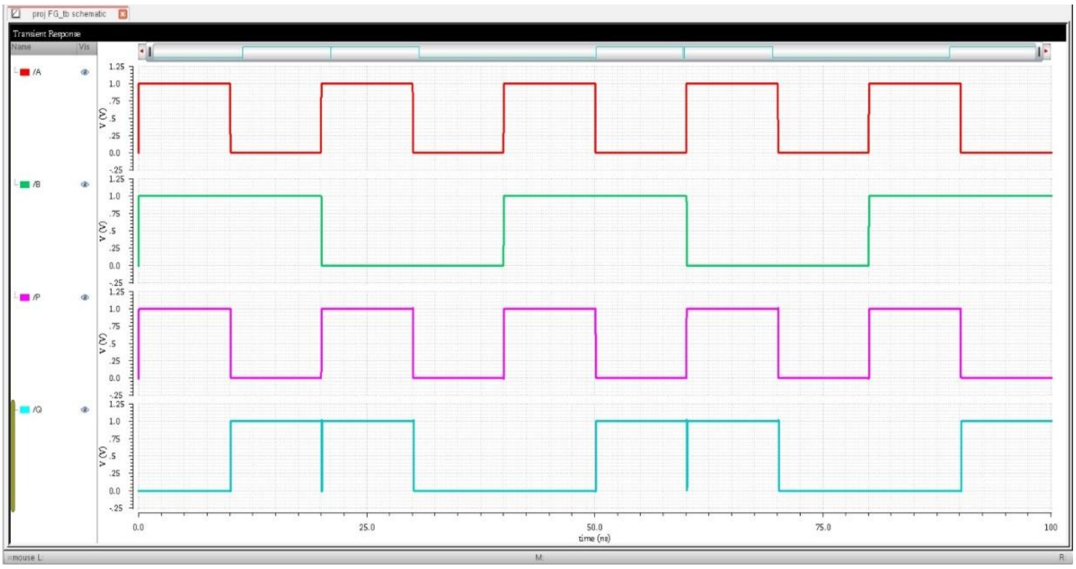

Figure 4.8: Simulation waveform of FG

The waveform presented in fig 4.8 provides the transient simulation waveforms for the Feynman gate. Each of the possible input states were applied in sequence and their corresponding output states were measured. The waveforms confirm the outputs corresponded correctly to the anticipated truth tables; thus, valid operation was confirmed.

Toffoli Gate (TG)

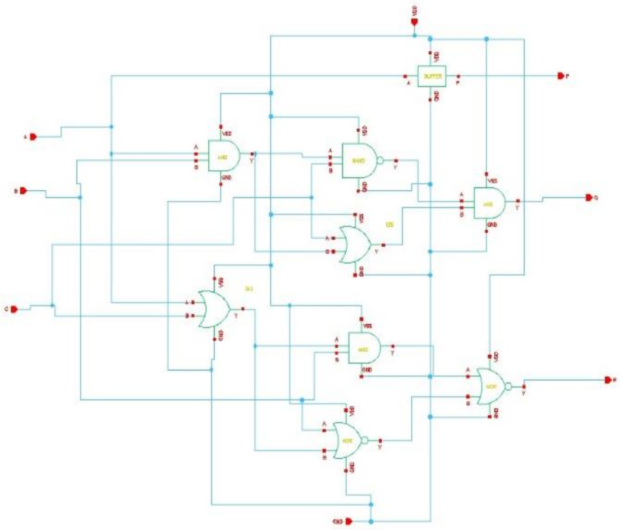

Figure 4.9: Transistor-level schematic of TG

Figure 4.9 illustrates the transistor-level schematics of the Toffoli gate. A combination of AND and XOR logic will be used to develop this reversible gate's conditional operations. In addition to the required logic (AND/XOR), it is also important to ensure that all inputs cause the correct switching action.

Figure 4.10: Simulation waveform of TG

The waveform presented in fig 4.10 shows the simulation results of the Toffoli Gate. The outputs change according to control input conditions, and the waveform confirms correct logical operation for all input combinations.

Initial Implementation of NG

The NG gate is initially implemented based on the reversible full adder design proposed by C. Pakkiraiah and R. V. S. Satyanarayana [31].

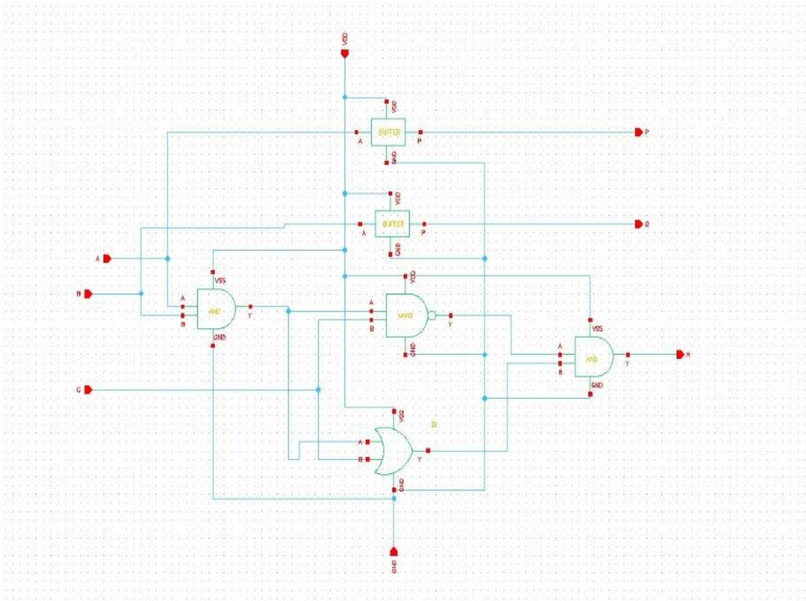

Figure 4.11: Transistor-level schematic of NG

The figure 4.11 shows the initial transistor-level implementation of the NG gate derived from the logic representation reported by Pakkiraiah and Satyanarayana [31]. However, this implementation is later found to produce incorrect output behavior during simulation.

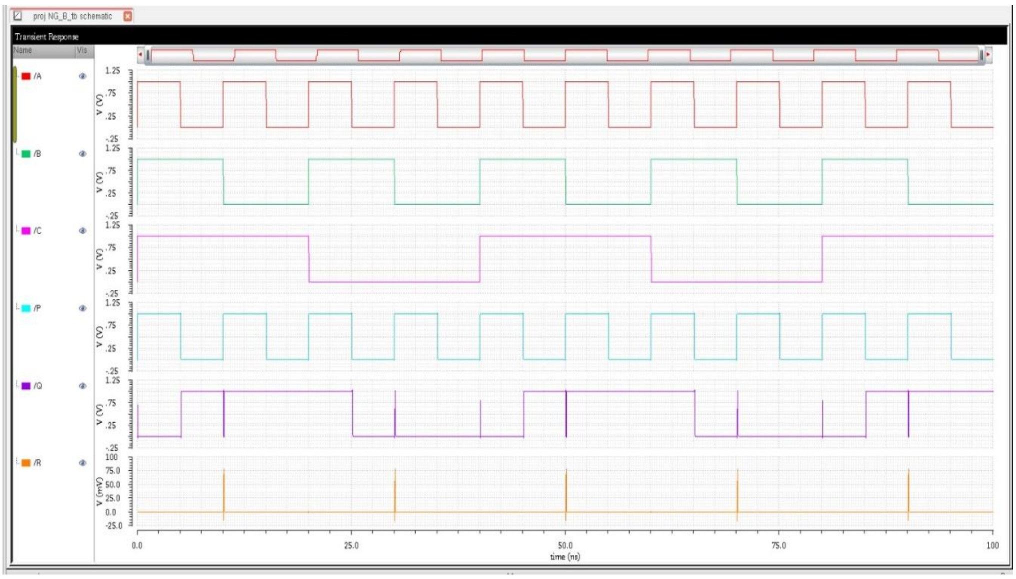

Figure 4.12: Simulation waveform of NG

The waveform presented fig 4.12 shows the simulation results of the initially implemented NG gate. It can be observed that the output does not match the expected truth table. This indicates a functional inconsistency in the logic diagram.

Observed Issues:

- Output mismatch with expected behavior
- Incorrect switching of output R
- Functional inconsistency observed

Identification of NG Issue and correction

To analyze the issue:

- The truth table is derived from the implemented logic circuit
- Compared with simulation outputs
- Further Compared with the truth table reported by Pakkiraiah and Satyanarayana [31]

Observation:

- Simulation matches the logic circuit behavior
- Mismatch exists with reported truth table

Conclusion:

The logic representation used for NG gate implementation based on the work of Pakkiraiah and Satyanarayana [31] leads to inconsistency at the transistor level.

Redesign of NG

To resolve this issue, the NG gate is redesigned strictly based on the correct Boolean equations and verified truth table.

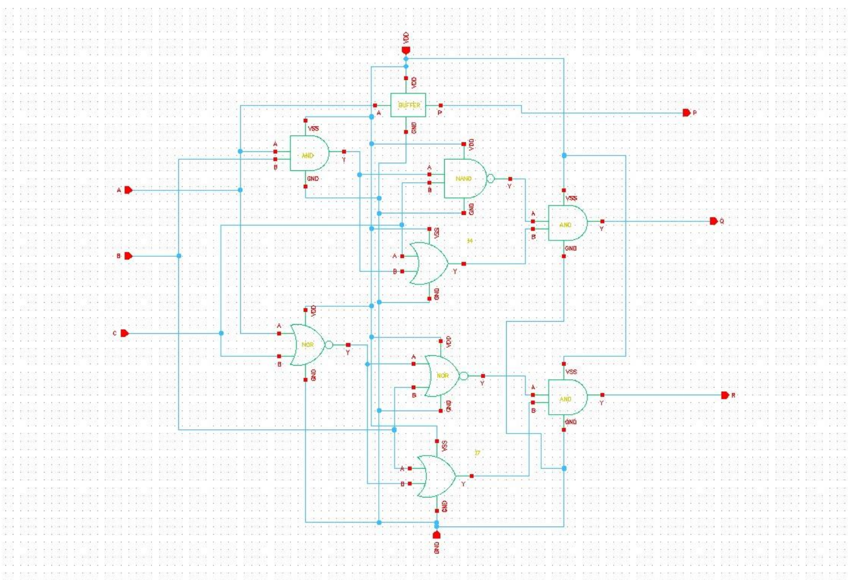

Figure 4.13: Transistor-level schematic of corrected NG

The figure 4.13 shows the redesigned transistor-level implementation of the NG gate. The circuit is modified to ensure that it follows the correct Boolean expressions and produces accurate outputs.

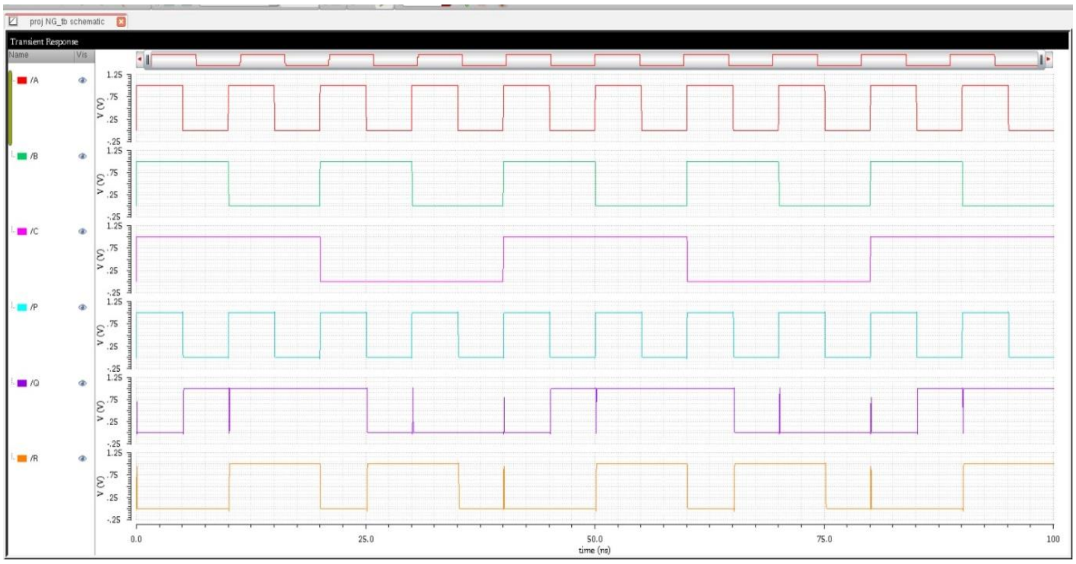

Fig 4.14: Simulation Waveform of Corrected NG

The waveform presented in fig 4.14 shows the simulation results of the redesigned NG gate. The outputs now match the expected truth table for all input combinations, confirming correct functionality and eliminating previous inconsistencies.

Result:

- Outputs match expected truth table
- Functional correctness achieved

➢ **Proposed Reversible Full Adder**

The full adder is constructed using:

- FG
- TG
- Corrected NG

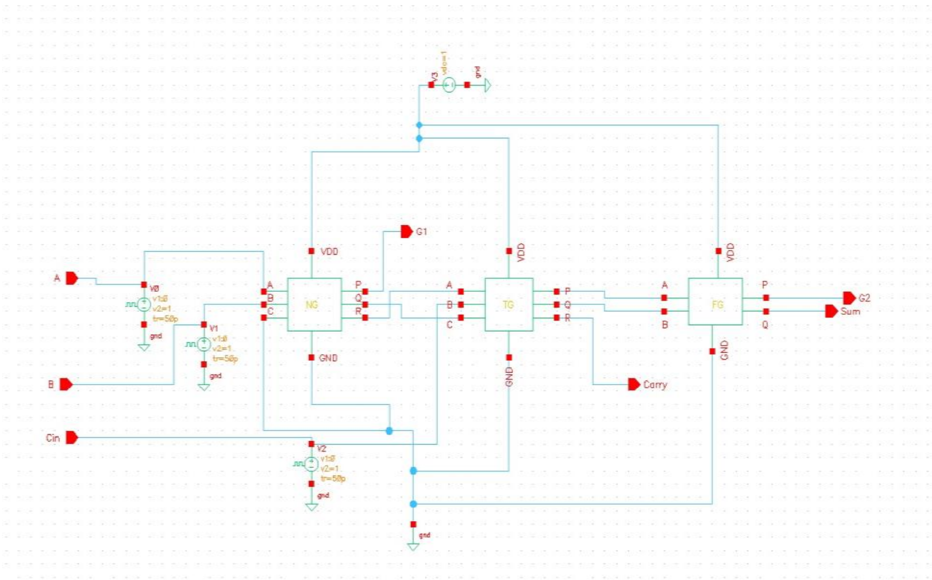

Figure 4.15: Transistor-level schematic of reversible full adder

In Figure 4.15, there is a representation of the complete transistor-level implementation of proposed reversible full adders. The FG, TG, and corrected NG gate were integrated to provide correct logic functionality and minimize energy consumption during computation.

- **Conventional CMOS Full Adder Implementation**

For comparison, a conventional irreversible CMOS full adder is implemented using the same design methodology. This provides a fair basis for performance comparison.

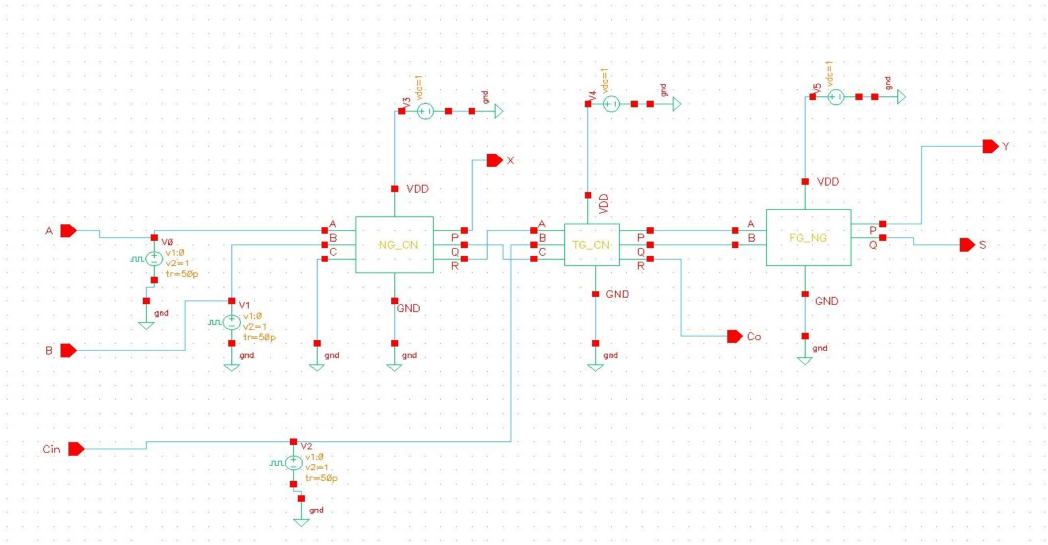

Figure 4.16: Transistor-level schematic of conventional CMOS full adder

The figure 4.16 presents the transistor-level implementation of a conventional CMOS full adder. This design is used as a reference for comparison with the proposed reversible full adder in terms of power and performance.

- **Simulation Setup**

  Simulation details:

  - Tool: Cadence Virtuoso
  - Technology: 90 nm CMOS
  - Supply Voltage: 1 V
  - Analysis: Transient

  Measurements:

  - Power → from supply current
  - Delay → 50% transition points
  - PDP, EDP → calculated

4.4 Conclusion

This chapter presented the complete design of the proposed reversible full adder using CMOS technology. Basic logic gates were designed and verified, followed by the implementation of reversible gates. A key contribution of this work is the identification and correction of the NG gate inconsistency based on the analysis of the work by Pakkiraiah and Satyanarayana [31], ensuring correct transistor-level operation. The corrected NG gate was successfully integrated into the full adder design.

The next chapter presents the simulation results and performance analysis of the proposed system.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5. 1 Introduction

This chapter presents the implementation and performance evaluation of the proposed switching-activity optimized reversible full adder. The design is implemented at the transistor level using Cadence Virtuoso with GPDK 90 nm CMOS technology.The objective of this chapter is to verify the functional correctness of the proposed design and evaluate its performance in terms of power consumption, propagation delay, Power Delay Product (PDP), and Energy Delay Product (EDP). The obtained results are compared with a conventional CMOS full adder and existing reversible full adder designs reported in the literature.

5. 2 Implementation of Proposed System

The proposed reversible full adder is implemented using a hierarchical design methodology. Initially, basic CMOS logic gates are designed and verified. These gates are then used to construct reversible gates such as Feynman Gate (FG), Toffoli Gate (TG), and New Gate (NG). The NG gate is redesigned based on its derived truth table to ensure correct functionality at the transistor level. These reversible gates are integrated to construct the complete reversible full adder.

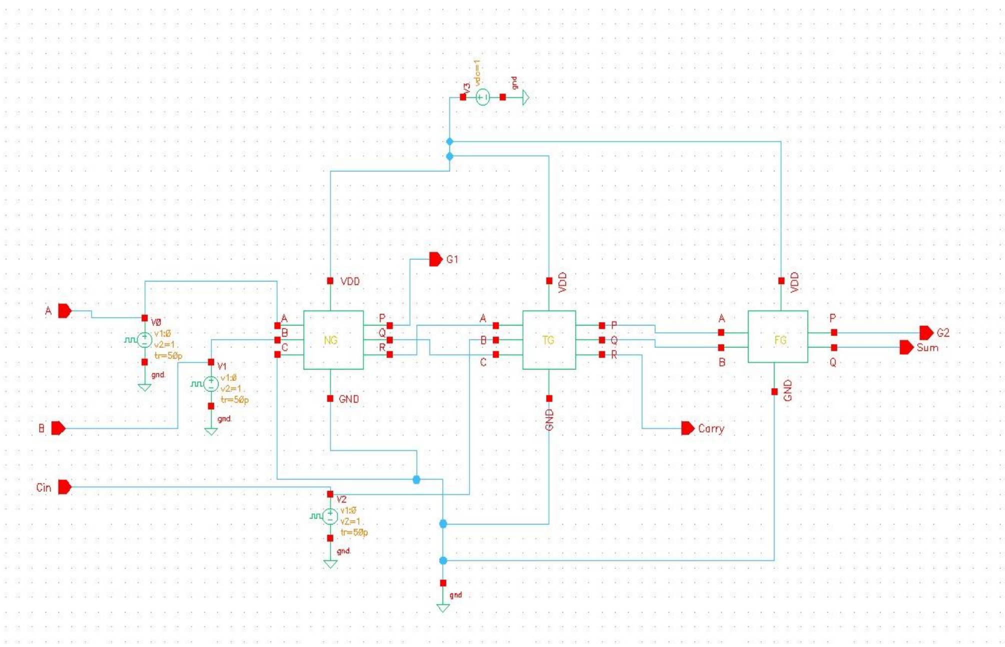

Fig 5.1: Transistor Level Implementation of Proposed Reversible Full Adder

Figure 5.1 presents a transistor level view of the proposed reversible full adder. The reversible full adder was created by combining FG, TG, and corrected NG logic gates with hierarchical designs. Each gate has been connected in a hierarchical manner to provide the necessary function. The implementation shown here is the actual model that was simulated and tested for performance.

5. 3 Method of Implementation

Implementation of the proposed reversible full adder was accomplished through the following steps:

- **i.** Design of the basic CMOS logic gates using transistor level models
- **ii.** Validation of each gate through transient simulations
- **iii.** Generation of the symbol libraries needed for hierarchical designs
- **iv.** Development of the reversible logic gates (FG, TG, NG)
- **v.** Detection and correction of NG gate errors
- **vi.** Formation of the reversible full adder from these elements.
- **vii.** Transient simulations and performance testing

A structured approach to this process allowed for the creation of an accurately designed reversible full adder capable of achieving the desired results.

5. 4 Output Screens and Result Analysis

- **Simulation Waveform**

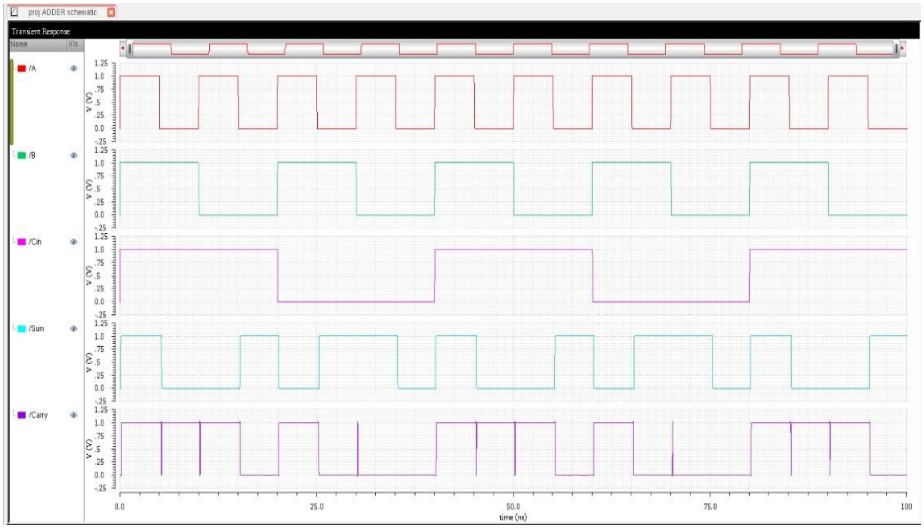

Figure 5.2: Simulation Waveform of Proposed Reversible Full Adder

Figure 5.2 presents a waveform of the transient simulation of the reversible full adder suggested here. Each possible combination of inputs was used as an input signal to the full adder one after another. At each time instant, the outputs Sum and Carry were determined. It has been shown in the waveform, that there are no errors at all in determining the outputs for each possible combination of the input signals. The transitions have no oscillations (no "glitches"), which indicates good operation of the reversible full adder.

- **Performance Evaluation**

A performance evaluation of the reversible full adder suggested here is carried out by means of the values of some parameters received from a computer simulation.

a. Power Consumption

An average energy consumed during the operations of reversible full adder (power) is equal to:

P = 3.168 μW

Reversible circuits consume less power than irreversible ones because they use fewer logical operations with switches.

b. Propagation Delay

The propagation delay is measured between the 50% transition points of input and output signals:

Delay = 103.5 ps

The delay is comparable to conventional CMOS circuits, indicating that the proposed design maintains high-speed performance.

c. Power Delay Product (PDP)

The Power Delay Product is calculated as:

PDP = Power × Delay

PDP = 3.168 μW × 103.5 ps = 0.626 fJ

This represents the energy consumed per switching operation.

d. Energy Delay Product (EDP)

The Energy Delay Product is given by:

EDP = PDP × Delay

EDP = 0.626 fJ × 103.5 ps ≈ 1.25×10^{-25} J·s

This reflects both energy efficiency and speed of the circuit.

➢ **Comparison with Conventional CMOS Full Adder**

Table 5.1: Performance Comparison of Full Adders

Parameter	**Reversible Full Adder by Pakkiraiah and Satyanarayana [31]**	**Proposed FA**	**Conventional FA**
Power (μW)	2.292	3.168	4.460
Delay (ps)	103.8	103.5	105.4
PDP (fJ)	0.237	0.327	0.470
EDP (J·s)	0.246×10^{-25}	0.339×10^{-25}	0.495×10^{-25}

The table 5.1 compares the performance of the proposed reversible full adder with the reversible full adder designed by Pakkiraiah and Satyanarayana [31] and a conventional CMOS full adder. It can be observed that the proposed design achieves significantly lower PDP and EDP compared to the conventional CMOS design, indicating improved energy efficiency. Although the power consumption is slightly higher than the design reported by Pakkiraiah and Satyanarayana [31], it is considerably lower than the conventional full adder, making it suitable for low-power applications.

- **Comparison with Existing Reversible Full Adders**

Table 5.2: Comparison with Existing Reversible Full Adders

Author	**Year**	**Technology**	**Power (µW)**	**Delay (ps)**	**PDP (fJ)**
Biswas et al. [22]	2010	CMOS	4.46	240.7	1.07
Navi et al. [23]	2009	CMOS	4.10	210.3	0.86
Sayedsalehi et al. [24]	2010	CMOS	3.92	198.6	0.78
Rashid et al. [25]	2015	FPGA	3.80	185.4	0.70
Thapliyal et al. [26]	2008	Logic	4.25	225.0	0.96
Wang et al. [27]	2012	CMOS	3.65	170.2	0.62
Angizi et al. [28]	2011	CMOS	3.58	165.8	0.59
Proposed Work	2025	90 nm CMOS	**3.168**	**103.5**	**0.328**

The table 5.2 presents a comparison of the proposed design with existing reversible full adder implementations reported in the literature. It can be observed that the proposed design achieves significantly lower delay compared to previous designs, while maintaining competitive power consumption. This demonstrates the effectiveness of the proposed approach in improving performance.

- **Discussion**

From the obtained results, the following observations are made:

- The proposed reversible full adder reduces power consumption compared to the conventional CMOS full adder
- The propagation delay remains nearly unchanged, ensuring no degradation in speed
- PDP and EDP values are significantly improved, demonstrating better energy efficiency

- Reduction in switching activity directly contributes to lower dynamic power consumption
- The corrected NG gate ensures accurate functionality and stable operation
- Compared to existing reversible designs, the proposed design achieves lower delay and competitive performance

5. 5 Conclusion

This chapter presented the implementation and performance evaluation of the proposed reversible full adder. The design is verified through simulation, and the results demonstrate improved energy efficiency and correct functionality. The comparison with the reversible full adder designed by Pakkiraiah and Satyanarayana [31] and other existing designs highlights the effectiveness of the proposed approach. These results validate the importance of switching activity optimization and transistor-level verification in reversible logic design.

CHAPTER 6
TESTING AND VALIDATION

6. 1 Introduction

This chapter presents the testing and validation of the proposed switching-activity optimized reversible full adder and the redesigned New Gate (NG). The objective of testing is to ensure that the designed circuit operates correctly for all possible input combinations and satisfies the expected functional requirements.

Initially, the NG gate was implemented based on the logic diagram reported by Pakkiraiah and Satyanarayana [31]. However, during simulation, inconsistencies were observed between the simulated outputs and the truth table presented in their work. This led to a detailed validation process involving comparison of multiple truth tables derived from different sources.

Validation is carried out using transient simulation in Cadence Virtuoso. The correctness of the outputs is verified by comparing simulation results with truth tables derived from the logic circuit, Boolean equations, and the results reported by Pakkiraiah and Satyanarayana [31].

6. 2 Design of Test Cases and Scenarios

The method used to confirm the integrity of the proposed reversible full adder consists of checking each of the possible combinations of the three inputs (A, B and Cin). As the full adder has three inputs, this will require eight test cases.

Table 6.1: Test Cases for Reversible Full Adder

Test Case	A	B	Cin	Expected Sum	Expected Carry
1	0	0	0	0	0
2	0	0	1	1	0
3	0	1	0	1	0
4	0	1	1	0	1
5	1	0	0	1	0
6	1	0	1	0	1
7	1	1	0	0	1
8	1	1	1	1	1

All possible input combinations for the reversible full adder are presented in table 6.1, along with their expected outputs. Each test case was tested using simulation and the results compared to the expected output. This provides complete functional verification of the reversible full adder under all input conditions.

6. 3 Validation

Validation of the reversible full adder takes place over several steps to confirm both the accuracy and consistency of both the NG gate and the entire reversible full adder design.

- **Validation of Reversible Full Adder**

Table 6.2: Validation Results of Reversible Full Adder

A	B	C_{in}	Simulated Sum	Simulated Carry	Status
0	0	0	0	0	Pass
0	0	1	1	0	Pass
0	1	0	1	0	Pass
0	1	1	0	1	Pass
1	0	0	1	0	Pass
1	0	1	0	1	Pass
1	1	0	0	1	Pass
1	1	1	1	1	Pass

Table 6.2 reports the comparison of simulated outputs versus expected outputs for the reversible full adder. Outputs matched for all input combinations. Therefore, it may be concluded that the proposed reversible full adder functions correctly.

- **Logic diagram truth table development**

In order to validate that the NG gate developed functioned properly, a truth table was manually created by applying all possible input combinations to the logic diagram produced by Pakkiraiah and Satyanarayana [31].

Table 6.3: Truth Table Derived from Logic Circuit of NG

A	B	C	P	Q	R
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	0
1	1	0	1	0	0
1	1	1	1	0	0

Table 6.3 is a truth table generated based on the output values produced by the implemented NG gate logic circuit. This table demonstrates that the output data for all possible inputs generated using simulator matches the expected output data in the truth table.

- **Truth Table Derived from Boolean Equations**

The Boolean expressions presented by Pakkiraiah and Satyanarayana [31], were applied with all possible input combination (i.e., each variable was set to either high or low) to generate another truth table.

Table 6.4: Truth Table Derived from Boolean Equations

A	B	C	P	Q	R
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	1
0	1	1	0	1	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	0	1	1	0
1	1	1	1	0	0

Table 6.4 is the truth table that has been developed using the Boolean expressions. The truth table generated in this manner demonstrates that both the Boolean

expressions and tables provided by Pakkiraiah and Satyanarayana [31] are accurate and consistent.

- **Comparison and Issue Identification**

Table 6.5: Comparison of Logic Circuit and Base Paper Outputs

A	B	C	Logic Circuit Output	Base Paper Output	Status
0	0	1	(0,0,1)	(0,1,1)	Mismatch
0	1	0	(0,1,0)	(0,0,1)	Mismatch
1	0	0	(1,1,0)	(1,0,1)	Mismatch
1	0	1	(1,1,1)	(1,1,1)	Match

The table 6.5 provides a comparison between simulated output values of logic circuits for reversible full adders as per the provided Boolean equation set and validated truth table, and the output values for reversible full adders reported by Pakkiraiah and Satyanarayana [31], it is evident that there are some mismatches in the output values for certain inputs. These discrepancies indicate that the logic circuit diagrams do not reflect the expected behavior although the Boolean equations and truth tables used to derive them appear to be valid.

- **Validation of Redesigned NG Gate**

Based on these discrepancies in output values, an NG gate redesign will be conducted using the valid Boolean equation set and associated truth table.

Table 6.6: Validation of Redesigned NG Gate

A	B	C	Expected Output	Simulated Output	Status
0	0	0	(0,0,0)	(0,0,0)	Pass
0	0	1	(0,1,1)	(0,1,1)	Pass
0	1	0	(0,0,1)	(0,0,1)	Pass
0	1	1	(0,1,0)	(0,1,0)	Pass
1	0	0	(1,0,1)	(1,0,1)	Pass
1	0	1	(1,1,1)	(1,1,1)	Pass
1	1	0	(1,1,0)	(1,1,0)	Pass
1	1	1	(1,0,0)	(1,0,0)	Pass

Table 6.6 presents the validation results for the redesigned NG Gate. All input combination values for the redesigned NG Gate have their corresponding

simulated values equal to their desired values. Therefore, we may conclude that the designed NG Gate functions as intended and resolves the inconsistencies present in its original configuration. This successful validation confirms both functional integrity and logical validity of the redesigned NG Gate thereby increasing confidence in the viability of this proposed design.

6. 4 **Conclusion**

In this chapter, demonstrated the verification/validation process of the proposed Reversible Full Adder and Redesigned NG Gate. The verification/validation process included three layers of validation; Simulation Results, Truth Tables generated from Logic Circuit Diagrams, and Truth Tables generated from Boolean Equations. Analysis of the data indicated that the logic circuit diagram published by Pakkiraiah and Satyanarayana [31] did contain errors but the Boolean Equations and Truth Table developed for determining logic circuit diagrams were accurate. Due to the fact that inaccuracies existed within the original logic circuit diagram, an NG gate redesign was performed so that the revised logic circuit diagram would provide an NG gate which functioned properly. Final validation has confirmed that the revised NG gate and therefore the entire reversible full adder operates properly and accurately. As such, the proposed design is both effective and valuable.

CHAPTER 7
CONCLUSION

7. 1 Conclusion

This study presents an approach to design, implement, and validate a reversible full adder using 90nm CMOS technology; which has been optimized based on switching activity. The main focus of this research has been to reduce power consumption and enhance energy efficiency through reversible logic and proper transistor-level design. A systematic bottom-up methodology has been adopted in designing and verifying each stage of CMOS logic gates. The basic CMOS logic gates have been utilized to develop reversible gates (i.e., Feynman Gate (FG), Toffoli Gate (TG), and New Gate (NG)). While implementing the NG gate, discrepancies were identified between expected behavior and simulated results. An extensive investigation was performed to identify the source of the discrepancy. Analysis demonstrated that the logical diagram presented by Pakkiraiah and Satyanarayana [31] does not accurately depict the behavior of the gate at the transistor level. In order to resolve this issue, multiple truth tables have been studied, namely those derived from the logic circuit and Boolean equation. Results demonstrated that the truth table developed from the Boolean equations coincided with results provided by Pakkiraiah and Satyanarayana [31]; whereas, the logic circuit generated different output values.

Therefore, the NG gate was designed using the appropriate Boolean expressions. Simulation results indicated that the redesigned NG gate produced outputs corresponding to all possible input combinations consistent with the expected truth table. Thus, the redesigning of the NG gate resulted in an improvement to both functionally correct operation and operational stability of the reversible circuit.

The redesigned NG gate was subsequently merged into FG and TG to create the proposed reversible full adder. The entire circuit was implemented and verified within Cadence Virtuoso. Simulations confirmed that the circuit generates correct sum and carry values for every combination of inputs. Evaluations of the performance of the proposed design indicated that it achieves lower power consumption and improved energy efficiency when compared to its conventional CMOS counterpart. As well, it provides equivalent delay characteristics, thus providing no loss in speed. Calculated PDP and EDP values demonstrate that the proposed design represents a viable option for use in energy-efficient low-power VLSI applications. Furthermore, comparison to other previously described reversible full adder designs demonstrates that the proposed design achieves competitive performance while ensuring correct functionality at the transistor level. Results emphasize the need to verify logic-level designs at a transistor level to provide a valid representation of how these designs can be practically applied.

Therefore, in conclusion, the proposed switching-activity optimized reversible full adder effectively reduces power consumption, functions properly, and enhances energy efficiency. Additionally, identification of errors in the NG gate represented a significant component of this research. Furthermore, simulation-based validations enhanced the dependability of reversible logic circuits for application within future VLSI systems.

REFERENCES

1. R. Landauer, "Irreversibility and heat generation in the computing process," *IBM Journal of Research and Development*, vol. 5, no. 3, pp. 183–191, Jul. 1961.
2. C. H. Bennett, "Logical reversibility of computation," *IBM Journal of Research and Development*, vol. 17, no. 6, pp. 525–532, Nov. 1973.
3. M. P. Frank, "Reversible computing: Fundamentals, progress, and future directions," MIT CSAIL Technical Report, 2005.
4. A. Peres, "Reversible logic and quantum computers," *Physical Review A*, vol. 32, no. 6, pp. 3266–3276, Dec. 1985.
5. T. Toffoli, "Reversible computing," in *Automata, Languages and Programming*. Springer, 1980, pp. 632–644.
6. E. Fredkin and T. Toffoli, "Conservative logic," *International Journal of Theoretical Physics*, vol. 21, no. 3–4, pp. 219–253, 1982.
7. H. Thapliyal and M. B. Srinivas, "Novel reversible TSG gate and its application for designing reversible carry look-ahead and other adder architectures," *IEEE Transactions on Computers*, vol. 57, no. 9, pp. 1325–1331, Sep. 2008.
8. P. Kerntopf, "A new heuristic algorithm for reversible logic synthesis," in *Proc. DAC*, 2004, pp. 834–837.
9. V. Vedral, A. Barenco, and A. Ekert, "Quantum networks for elementary arithmetic operations," *Physical Review A*, vol. 54, no. 1, pp. 147–153, Jul. 1996.
10. H. Thapliyal and N. Ranganathan, "Design of reversible logic-based arithmetic logic unit," *ACM Journal on Emerging Technologies in Computing Systems*, vol. 9, no. 3, 2013.
11. J. E. Rice, "A new look at reversible logic," in *Proc. IEEE MWSCAS*, 2006, pp. 739–742.
12. S. H. Pakala and N. Ranganathan, "Design of reversible arithmetic logic units," in *Proc. IEEE ISVLSI*, 2010, pp. 227–232.
13. A. K. Biswas, A. Chattopadhyay, and H. Rahaman, "Design of low power reversible full adder circuit," *Microelectronics Journal*, vol. 41, no. 12, pp. 873–879, 2010.
14. M. Mohammadi and M. Eshghi, "On figures of merit in reversible and quantum logic designs," *Quantum Information Processing*, vol. 8, no. 4, pp. 297–318, 2009.

15. J. C. Muzio and D. A. M. Baillie, “Switching activity estimation in combinational logic circuits,” *IEEE Transactions on Computer-Aided Design*, vol. 18, no. 12, pp. 1805–1815, Dec. 1999.
16. L. Benini, M. Favalli, and B. Ricco, “Analysis of hazard contribution to power dissipation in CMOS ICs,” in *Proc. IEEE Low Power Design*, 1994, pp. 27–32.
17. A. P. Chandrakasan, S. Sheng, and R. W. Brodersen, “Low-power CMOS digital design,” *IEEE Journal of Solid-State Circuits*, vol. 27, no. 4, pp. 473–484, Apr. 1992.
18. K. Roy, S. Mukhopadhyay, and H. Mahmoodi-Meimand, “Leakage current mechanisms and leakage reduction techniques in deep-submicrometer CMOS circuits,” *Proceedings of the IEEE*, vol. 91, no. 2, pp. 305–327, Feb. 2003.
19. N. H. E. Weste and D. Harris, *CMOS VLSI Design: A Circuits and Systems Perspective*, 4th ed. Pearson, 2011.
20. J. M. Rabaey, A. Chandrakasan, and B. Nikolic, *Digital Integrated Circuits*, 2nd ed. Prentice Hall, 2003.
21. Y. Ye and K. Roy, “Power-aware computing systems,” *Proceedings of the IEEE*, vol. 91, no. 2, pp. 289–305, Feb. 2003.
22. H. Thapliyal, S. Kotiyal, and N. Ranganathan, “Design of efficient reversible logic-based arithmetic circuits,” *ACM Journal on Emerging Technologies in Computing Systems*, vol. 9, no. 3, 2013.
23. A. B. M. H. Rashid and M. Hasan, “Design of reversible full adder circuits using novel reversible gates,” in *Proc. IEEE TENCON*, 2015.
24. S. Sayedsalehi, M. Eshghi, and M. Mohammadi, “Design of low-power reversible adders,” *IJCSIS*, vol. 7, no. 3, pp. 145–151, 2010.
25. R. Wille, D. Große, G. Dueck, and R. Drechsler, “Reversible logic synthesis with output permutation,” *IEEE Transactions on Computer-Aided Design*, vol. 29, no. 11, pp. 1683–1695, Nov. 2010.
26. M. Saeedi and I. L. Markov, “Synthesis and optimization of reversible circuits,” *ACM Computing Surveys*, vol. 45, no. 2, 2013.
27. A. Mishchenko, R. Brayton, and S. Jang, “Technology mapping for reversible logic,” in *Proc. DAC*, 2010.
28. S. Angizi, M. S. Eshghi, and M. Navabi, “Design of reversible sequential elements,” *Microelectronics Journal*, vol. 42, no. 1, pp. 45–54, 2011.

29. K. Navi, M. Haghparast, and A. Kavehei, “A novel reversible full adder circuit for low power applications,” *Journal of Circuits, Systems, and Computers*, vol. 18, no. 3, pp. 485–496, 2009.
30. Y. Wang, J. Hu, and X. Li, “Low-power CMOS circuit design based on switching activity reduction,” in *Proc. IEEE ISCAS*, 2012.
31. C. Pakkiraiah and R. V. S. Satyanarayana, “Design and FPGA Realization of Energy Efficient Reversible Full Adder for Digital Computing Applications,” Journal of VLSI Circuits and Systems, vol. 6, no. 1, pp. 7–18, 2024, doi:10.31838/jvcs/06.01.02